PLANE TABLE

• Graphical method

• No field book required

• No office work necessary

• Field work and plotting proceed simultaneously

• Fastest method in conventional surveying

• Less accurate

• Preferred for large areas (small scale maps)

Advantages:

• Minimal chance of omitting observations

• Simple instrument

Disadvantages:

• Difficult to replot the drawing at different scales

• Topical instrument (cannot be used in wet conditions)

Temporary adjustments of plane table

Centring

• Done with a U-frame instrument with plumb bob

• In plane table surveying, centring and levelling are done simultaneously

Levelling

• Done with a spirit level or flat bubble

Orientation

• Positioning the instrument so that its position is parallel to its previous positions

• Ensuring all lines already plotted are parallel to corresponding lines on the ground

- Levelling and centring done simultaneously

Methods of Orientation:

By Compass

• Less accurate

• Cannot be used in areas affected by local attraction

By back sighting

• Accurate method

• Commonly used method

Instruments and accessories used in plane table surveying

1) Plane table - 3 types:

a. Traverse table

• Levelling done by leg adjustment

b. Johnson's table (45x60 cm and 60x75 cm)

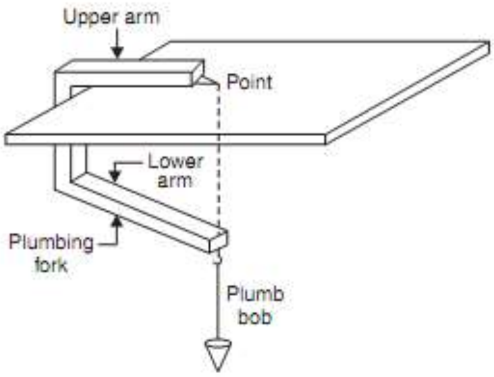
• Levelling done by ball and socket mechanism

c. Coast Survey table

• Levelling done by 3 foot screws in levelling head

• Used for accurate work

2) U-frame with plumb bob:



• Used for centring

• Horizontal arm or upper arm with pointer

• Inclined or lower arm with hook

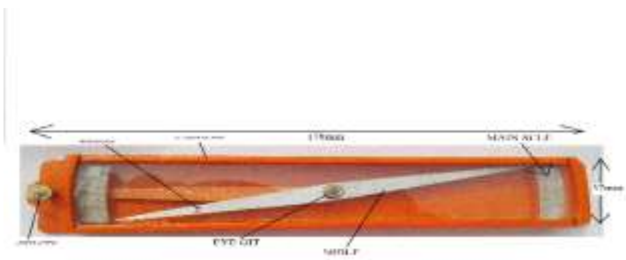
3) Spirit level:

• Flat based

• Used for levelling plane table.

4) Compass :

• Trough compass



• Circular box compass

• Line compass

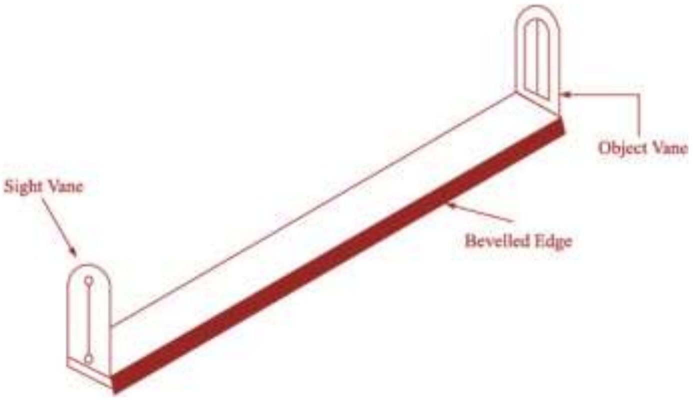
• Tubular compass

5) Alidade :

• Used for sighting

2 types :

Plain Alidade:



• Typically 50cm long

• Made of brass or gun metal

• Has an eye vane with a narrow slit and an object vane with a wider slit

• Has a vertical hair or wire

• The working edge is the fiducial edge

• The edge with measurements marked is where the measurements are taken

Telescopic Alidade:

• Provides increased accuracy in bisecting

• Increases the range of bisecting

• Provides the inclination of the line of sight

• Allows for tacheometry (the technique of surveying by measuring horizontal distances and differences in elevation)

Methods of Plane Tabling

1) Radiation

• Used to plot nearby and accessible points

• Requires only one instrument station as a reference

• Can cover the entire area from a single instrument station

2) Intersection

• Used when stations are inaccessible

• Also known as graphical triangulation

• Requires two instrument stations as references

3) Traversing

• Involves moving through the surveying area while collecting details

• Requires multiple instrument stations

4)Resection

• To locate position of instrument station with respect to plotted reference points

a) Two-point problem: Not reliable, with inaccessible reference points

b) Three-point problem: Accurate

<u>Three-point problem</u>

Methods:

1) Mechanical (tracing paper/tracing cloth method)

2) Bessel's graphical solution

3) Lehmann's method/trial and error method/triangle of error method

Lehmann's Method:

• If the orientation is wrong, the three resectors will form a small triangle called the "triangle of error"

• If the orientation is correct, the three resectors will converge at the same point (instrument station)

Lehmann's Rule:

Explains how much and in which direction the table should be rotated to solve the triangle of error, using a trial and error method.

Strength of Fix is good if:

• The instrument station is located within the great triangle (the imaginary triangle formed by joining the three reference stations)

• The station is closer to the middle reference object

• One of the angles subtended at the station is small, and the other is large

Strength of Fix is poor if:

• The station is located on the circumference of the great circle (the imaginary circle passing through the three reference stations)